

1 How do you generate random numbers in Java Script?

- a) **Using math.random()**
- b) Using random()
- c) Using getRandom()
- d) Using generate random()

2 How do you create a multiline string in JavaScript?

- a) Using double quotes
- b) Using backticks**
- c) Using parentheses
- d) Using single quotes

3 What are the let const, and class keywords introduced in ES6? How do they differ from their ES5 counterparts?

- a) let for block-scoped variables, const for constants, class for classes**
- b) let for global variables, const for constants, class for functions
- c) let for functions, const for block-scoped variables, class for arrays
- d) let for objects, const for global variables, class for constants

4 What is jQuery and what is its main purpose in web development?

- A) A CSS framework
- B) A JavaScript library for simplifying HTML DOM manipulation, event handling, and animation**
- C) A backend framework
- D) A database management system

5 What is the purpose of the "alt" attribute in HTML image tags?

- a) It specifies the alignment of the image.
- b) It defines the size of the image.
- c) It provides alternative text for the image for accessibility.**
- d) It controls the animation of the image.

6 What is Bootstrap and what role does it play in web design?

- A) A JavaScript framework
- B) A CSS framework for building responsive and mobile-first websites**
- C) A database management system
- D) A server-side scripting language

7 Which CSS property is used to control the spacing between lines of text?

- a) text-spacing
- c) letter-spacing

b) line-spacing d) line-height

8 What does the "float" property in CSS do?

a) It controls the alignment of text within an element.

b) It removes an element from the normal flow of the document and positions it to the left or right of its containing element.

c) It makes an element visible or hidden.

d) It rotates an element clockwise or counterclockwise.

9 What does the "z-index" property in CSS control?

a) It controls the size of an element.

b) It controls the position of an element in the stacking order.

c) It controls the opacity of an element.

d) It controls the rotation of an element.

10 Which HTML tag is used to define a table row?

a) <tr> c) <th>

b) <td> d) <table>

11 Which keyword is used to declare a variable in JavaScript?

a) var c) v

b) variable d) let

12 Which operator is used for strict equality comparison in JavaScript?

a) == c) ===

b) != d) !==

13 Which function is used to output data to the console in JavaScript?

a) print() c) display()

b) log() d) console.log()

14 Which built-in method is used to convert a string to an integer in JavaScript?

a) parseInt () c) convert Toint()

d) string ToInt() b) toInteger()

15 What does the method querySelector() do in JavaScript?

a) It selects multiple elements based on a class name.

b) It selects the first element based on a CSS selector.

c) It selects an element based on its ID.

d) It selects all elements of a specific tag name.

16 How do VPNs (Virtual Private Networks) enhance intranet security and connectivity?

VPNs encrypt data, making it secure during transmission. They allow remote access to intranet resources by creating a secure, private connection over the internet.

17 How does an interactive website differ from static and dynamic websites?

- ❑ Static websites are fixed and show the same content to all users.
- ❑ Dynamic websites fetch content dynamically from a database based on user interactions.
- ❑ Interactive websites provide richer engagement, allowing user input and interactive elements like forms, animations, and real-time updates

18 Which code editor do you prefer for web design and development projects, and why?

Examples include VS Code for its flexibility, extensions, and debugging tools, or Sublime Text for its lightweight performance.

19 Can you explain the basic purpose of FTP (File Transfer Protocol) software in web development?

FTP is used to transfer files between a local computer and a web server, making it essential for uploading and managing website content.

20 How do you create HTML login form with submit button?

```
<form action="/submit-login" method="POST">
<label for="username">Username:</label>
<input type="text" id="username" name="username">
<label for="password">Password:</label>
<input type="password" id="password" name="password">
<button type="submit">Login</button>
</form>
```

21 When should I consider using SVG images?

Use SVGs for scalable graphics like icons and logos because they maintain quality at any resolution and are lightweight.

22 What is an ID selector in CSS?

An ID selector targets a specific element using its unique ID, prefixed with # in CSS.

23 How do you apply multiple CSS classes to a single HTML element?

```
<div class="class1 class2"></div>
```

24 How do you create functions in JavaScript?

```
function greet(name) {
  return `Hello, ${name}!`;
}
```

25 Can you explain the difference between function declarations and function expressions?

Function Declaration: Defined with function keyword and hoisted

Function Expression: Stored in a variable and not hoisted.

26 What are arrays and objects in JavaScript?

❑ Array: Ordered collection of elements.

Example: [1, 2, 3]

❑ Object: Key-value pairs.

Example: { name: "Alice", age: 25 }

27. What are arrays and objects in JavaScript?

❑ Array: Ordered collection of elements.

Example: [1, 2, 3]

❑ Object: Key-value pairs.

Example: { name: "Alice", age: 25 }

28 Can you explain the difference for loop, while loop and do while loop?

For loop: Executes a set number of times

While loop: Runs while a condition is true.

Do-while loop: Executes at least once, then checks the condition.

29 How do you handle conditional statement in java script?

Using if-else, switch, or ternary operators.

30 How do you create new Google Sheet is your Google Drive

❑ Open Google Drive.

❑ Click **New > Google Sheets**.

❑ Name and customize your sheet.

1. What tools and software do I need for web design?

Answer: You'll need a code editor (like VS Code or Sublime Text), a web browser (like Chrome or Firefox), graphic design software (like Figma or Adobe XD), and possibly a local server environment (like XAMPP or MAMP).

2. What are the types of networks?

Intranet

Extranet

Internet

3. Write the types of website.

Static

Dynamic

Interactive

4. Write 2 FTP clients' names.

FileZilla

CyberDuck

File Manager

5. What are the basic components of an HTML document?

Answer: An HTML document typically consists of the `<!DOCTYPE html>` declaration, `<html>`, `<head>`, and `<body>` tags. The `<head>` contains meta-information (like title and links to stylesheets), while the `<body>` contains the content displayed on the web page.

6. How do you create a hyperlink in HTML?

Answer: Use the `<a>` (anchor) tag. For example: `<a href="https://www.example.com">Visit Example</a>`. This creates a clickable link to the specified URL.

7. How can you add images to a webpage using HTML? Answer: Use the tag, specifying the src attribute for the image source and

the alt attribute for accessibility. For example: ``.

8. How can you link an external CSS file to your HTML document?

Use the `<link>` tag within the `<head>` section. For example:

```
<link rel="stylesheet" href="styles.css">
```

9. What is the purpose of the <!DOCTYPE html> declaration?

Answer: The `<!DOCTYPE html>` declaration defines the document type and version of HTML being used. It helps browsers render the page correctly and ensures standards compliance.

10. What is the purpose of the <meta> tag in HTML?

Answer: The `<meta>` tag provides metadata about the HTML document, such as character encoding, viewport settings for responsive design, and descriptions for SEO. For example: `<meta name="viewport" content="width=device-width, initial-scale=1.0">`.

11. What are the key differences between raster and vector graphics?

Answer: Raster graphics are made of pixels and are resolution-dependent (e.g., JPEG, PNG), while vector graphics are made of paths defined by mathematical formulas, making them resolution-independent (e.g., SVG, AI). Vectors can be resized without losing quality.

12. What file formats should I use for saving images?

Answer: Use JPEG for photos and web images due to its compression, PNG for images requiring transparency, and TIFF for high-quality prints. Use PSD or XCF for working files if you need to preserve layers for future editing.

13. What are the most popular image editing software options?

Answer: Popular options include Adobe Photoshop, GIMP (GNU Image Manipulation Program), Affinity Photo, Canva, and Adobe Lightroom. Each has unique features catering to different user needs.

14. What is CSS, and why is it important in web design?

Answer: CSS (Cascading Style Sheets) is a stylesheet language used to control the presentation of HTML elements. It allows designers to separate content from layout, making it easier to maintain and style websites consistently.

15. What are the different ways to apply CSS to a web page?

Answer: CSS can be applied in three ways: inline (using the style attribute in HTML elements), internal (within a <style> tag in the <head> section), and external (linking to an external .css file using the <link> tag).

16. What is the difference between classes and IDs in CSS?

Answer: Classes are reusable styles applied to multiple elements (using . notation), while IDs are unique styles applied to a single element (using # notation). IDs have higher specificity than classes in CSS.

17. What are CSS frameworks, and what advantages do they offer?

Answer: CSS frameworks are pre-prepared libraries that provide a foundation for designing websites. They offer advantages like faster development, consistent design patterns, and built-in responsiveness. Popular frameworks include Bootstrap, Tailwind CSS, and Foundation.

18. How does Bootstrap help in web design?

Answer: Bootstrap is a popular CSS framework that provides a grid system,

pre-defined components (like buttons, forms, and modals), and responsive design utilities. It simplifies the process of creating mobile-friendly layouts.

19.What are variables in JavaScript, and how do you declare them?

Answer: Variables store data values. In JavaScript, you can declare variables using `var`, `let`, or `const`. For example:

```
let name = "John"; // can be reassigned
```

```
const age = 30;
```

```
// cannot be reassigned
```

20.What are functions in JavaScript, and how do you create one?

Answer: Functions are reusable blocks of code designed to perform a specific task.

You can create a function using the function keyword:

```
function greet() {console.log("Hello, World!");  
}
```

21.How do you work with arrays in JavaScript?

Answer: Arrays store multiple values in a single variable. You can create an array using square brackets:

```
let fruits = ["apple", "banana", "cherry"];
```

```
console.log(fruits[1]); // outputs "banana"
```

22.What is the difference between `==` and `===` in JavaScript?

Answer: `==` is the equality operator that checks for value equality with type coercion,

while `===` is the strict equality operator that checks for both value and type equality.

It's generally recommended to use `===` to avoid unexpected behaviour.

1. What is meant by Website?

Ans: A website is a collection of interconnected web pages, typically hosted on a single domain name, and accessible via the Internet.

2. What are the most common communication Channels used for effective communication?

Ans: Face to Face, Phone Call, Social Media

3. What is an attribute?

Ans: an attribute refers to additional information or properties that can be added to HTML elements to modify their behavior or appearance.

4. What is meant by Internet Protocol (IP address)?

Ans: Unique Address of a device connected to the internet.

5. What does HTML stand for?

Ans. Hyper Text Markup Language.

6. How do you insert an image in HTML?

Ans:

7. How does computer Hardware differ from software?

Ans: Software: Needs Logical Instructions.

Hardware: Do not need Logical Instructions.

8. What is the purpose of the <head> section in an HTML document?

Ans: The <head> section in an HTML document is used to contain meta information about the document, such as the title, character set, styles, and links to external resources like CSS files.

9. How do you create a table in HTML?

Ans: <table>

<tr>

<th>Header 1</th>

<th>Header 2</th>

</tr>

<tr>

<td>Data 1</td>

<td>Data 2</td>

</tr>

</table>

10. .What is the difference between padding and margin in CSS?

Ans: Padding: Create space inside the elements border.

Margin: Create space outside the element's border.

11. How do you create a responsive layout in CSS?

Ans: Using Media Query

SB SIZAN12. What is an interactive website?

Ans: A website that can communicate with users and allows them for interaction.

13. What is JavaScript?

Ans. JavaScript is a high level programming language.

14. How do you declare a variable in JavaScript?

Ans: var, let, const.

```
var name = 'rahim';
```

```
let district = 'dhaka';
```

```
const fatherName = 'jashim';
```

15. When will a web developer's personal computer be called a local server?

Ans. When developers install Xampp on their personal computer.

16. What is the function of an FTP client?

Ans: The function of an FTP client is to transfer files between a local computer and a remote server over the File Transfer Protocol (FTP) network.

17. Which color mode is used for printing?

Ans: CMYK

18. Why is a block statement is Used in Java script?

Ans: A block statement in JavaScript is used to group multiple statements together into a single compound statement.

19. When interpreter encounters an empty statement, what will it do?

Ans: It will show a warning and skip the statement and will proceed to the next statement.

20. Write Down the correct sequence of HTML tags for starting a webpage.

Ans: <html> <head> <title> <body>

21. Write the tag used for inserting the largest heading in HTML.

Ans: <h1>

22. Write the tag used for inserting a line-break in HTML.

Ans:

23. How will you create a hyperlink in HTML?

Ans: Link Text

24. What is meant by cascading style sheets?

Ans: The short form of cascading style sheets is CSS is responsible for adding style on HTML web structure.

25. Write the correct syntax for referring to the external style sheet.

Ans: <link rel="stylesheet" type="text/css" href="styles.css">

26. The "function" and "var" are known as:

Ans: Keywords.

SB SIZAN27. Write the code in Javascript for getting an output 1.

Ans: console.log(1);

alert(1);

28. What will happen, if the following Javascript code is executed?

```
let number = 1;
while (number <= 9) {
  console.log(number);
  number++;
}
```

Ans: 1 - 9

29. List the data type supported by javascript.

Ans: Number

String

Boolean

Undefined

Null

Symbol

Object

30. What are the different types of lists in HTML?

Ans: Unordered List

Ordered List

31. What is the purpose of the <div> and tags?

Ans: The <div> tag is used for grouping block-level elements for styling and layout, while the tag is used for inline styling and grouping inline elements in HTML.

32. What are the different ways to apply CSS to a web page?

Ans: Inline, Internal, External.

33. How do you use CSS to change the color of text?

```
Ans: selector {
  color: red;
}
```

34. What is the difference between class and ID selectors in CSS?

Ans: Class selectors in CSS are denoted by a dot (.) and can be used multiple times on a page, while ID selectors are denoted by a hash (#) and should be unique within a page

35. How do you apply multiple CSS classes to a single HTML element?

Ans: By separating each class name with a space in the class attribute.

```
<div class="class1 class2 class3"></div>
```

36. What is the purpose of using <iframe>?

Ans: The purpose of using <iframe> in HTML is to embed another document in the current HTML document, typically used for displaying content from another source like a webpage and videos.

37. How do you create a comment in HTML?

Ans: <!-- Comments -->

38. How do you create a hyperlink that opens in a new tab?

Ans: By using target attribute and the value of the target attribute will be “_blank”

```
<a href="https://example.com" target="_blank"> Link text </a>
```

39. How do you apply a background image to an element using CSS?

Ans: using the background-image property with the URL of the image.

.element

```
{ background-image: url('image file path');  
}
```

40. How do you make a font bold using CSS?

Ans: By using font-weight property. And the value of this property will be 700.

41. How do you hide an element using CSS?

Ans: There are three ways we can hide an element. display: none; visibility: hidden; opacity: 0;

42. What are meta tags in HTML?

Ans: Meta tags in HTML provide metadata about the HTML document, such as character set, viewport settings, author, and keywords for search engines, but are not displayed on the web page itself.

43. How do you create a checkbox in an HTML form?

Ans: <input type="checkbox">

44. How do you create a hover effect in CSS?

Ans: using the ‘:hover’ pseudo-class

45. How to add a comment in CSS?

Ans: /*

commented text here

*/